# Successive Differentiation

**Def$^{n}$:**

The function $f'$ defined by the formula

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

is called the derivative of $f$ with respect to $x$. The process of finding a derivative is called the differentiation. The differentiation operation is often denoted by the symbol $\frac{d}{dx}[f(x)]$. Thus $\frac{d}{dx}[f[x]] = f'(x)$. The derivative $f'(x)$ gives the slope of the graph at $x$ and also the instantaneous rate of change of $f$ at $x$.

## Differentials:

Let $\Delta x = dx$ be an increment gives to $x$, then

$\Delta y = f(x+\Delta x) - f(x)$ is called the increment in $y = f(x)$. If $f(x)$ is continuous and a continuous first derivative in an interval, then

$$\Delta y = f'(x)\,\Delta x + \varepsilon \cdot \Delta x$$

$$= f'(x)\,\Delta x + \epsilon\, dx \qquad \text{——— (2)}$$

when $\epsilon \to 0$ as $\Delta x \to 0$

The expression $dy = f'(x)\,dx$ ——③ is called the differential of $y = f(x)$.

The quantity $dx$, called the differential of $x$ and $dy$ need not be small.

Because of definitions ① and ③, we often write

$$\frac{dy}{dx} = f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

It is emphasized that $dx$ and $dy$ are not the limits of $\Delta x$ and $\Delta y$ as $\Delta x \to 0$. Instead, $dy$ is a dependent variable determined from the independent variable $dx$ for a given $x$.

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# Successive Differentiation

## Some basic theorems :

**Th$^m$ :**

If $y = x^n$, $n \in \mathbb{N}$, then show that $y_n = n!$

**Proof :**

Given, $y = x^n$

Differentiating wrt $x$ we get

$$y_1 = nx^{n-1}$$

Again differentiating wrt $x$ we obtain

$$y_2 = n(n-1)x^{n-2}$$

Similarly, $y_3 = n(n-1)(n-2)x^{n-3}$

$$y_n = n(n-1)(n-2)\cdots 3.2.1\, x^{n-n}$$

$$= n(n-1)(n-2)\cdots 2.1.$$

$$= n!$$

(Proved)

**Th^m :**

If $y = x^m$ and $m \in N$, then show that

(a) $y_n = 0$, when $n > m$

(b) $y_n = \frac{m!}{(m-n)!} x^{m-n}$, when $n \leq m$.

**Proof : (a)**

Given, $y = x^m$

Therefore, using the previous th^m we obtain

$$y_m = m!$$

Thus, $y_n = 0$ if $n > m$.

**Proof : (b)**

Given, $y = x^m$

Diff wrt $x$ we get

$$y_1 = mx^{m-1}$$

Diff. again wrt $x$ we obtain

$$y_2 = m(m-1)x^{m-2}$$

Similarly,

$$y_3 = m(m-1)(m-2)x^{m-3}$$

- - - - .

$$y_n = m(m-1)(m-2)\cdots\{m-(n-1)\}x^{m-n} \text{ when } m \geq n$$

$$\Rightarrow y_n = m(m-1)(m-2)\cdots(m-n+1)x^{m-n}$$

$$= \frac{m(m-1)(m-2)\cdots(m-n+1)(m-n)!}{(m-n)!}x^{m-n}$$

$$= \frac{m(m-1)(m-2)\cdots(m-n+1)(m-n)(m-n-1)\cdots 3\,2\,1}{(m-n)!}x^{m-n}$$

$$= \frac{m!}{(m-n)!}x^{m-n}, \text{ when } m \geq n.$$

(Proved)

**Th$^m$:**

If $y = (ax+b)^n$ and $n \in \mathbb{N}$, then show that $y_n = n!\,a^n$.

**Proof:**

Given, $y = (ax+b)^n$

Diff. wrt $x$ we get.

$$y_1 = n(ax+b)^{n-1} \cdot a$$

Again diff wrt $x$ we have

$$y_2 = n(n-1)(ax+b)^{n-2} \cdot a^2$$

Similarly,

$$y_3 = n(n-1)(n-2)(ax+b)^{n-3} a^3$$

. . . . . .

$$y_n = n(n-1)(n-2)(n-3)\cdots 3.2.1 \cdot a^n$$

$$\therefore y_n = n! a^n$$

(Proved)

Th$^m$:

If $y = a^{bx+c}$, then show that $y_n = a^{bx+c}(\ln a)^n$.

Proof:

Given, $y = a^{bx+c}$ ——— (1)

Diff (1) wrt $x$ we get

$$y_1 = a^{bx+c} \cdot \ln a \cdot b$$

Again diff wrt $x$ we have

$$y_2 = a^{bx+c}(\ln a)^2 \cdot b^2$$

Similarly,

$$y_3 = e^{bx+c}(\ln a)^3 b^3$$

- - - - .

$$y_n = e^{bx+c}(\ln a)^n b^n$$

(Proved)

Th^m:

If $y = \ln(ax+b)$, then show that:

$$y_n = (-1)^{n-1}(n-1)!\,(ax+b)^{-n} a^n$$

Proof:

Given, $y = \ln(ax+b)$ ——— (1)

Diff. (1) wrt x we get.

$$y_1 = \frac{1}{ax+b} \cdot a = (ax+b)^{-1} \cdot a$$

Again diff wrt x we have

$$y_2 = (-1)(ax+b)^{-2} \cdot a^2$$

Similarly, $y_3 = (-1)(-2)(ax+b)^{-3} \cdot a^3$

- - - .

$$y_n = (-1)(-2)\cdots(-n+1)(ax+b)^{-n} \cdot a^n$$

$$= (-1)^{n-1}\left[1.2.3\cdots(n-1)\right](ax+b)^{-n} a^n$$

$$= (-1)^{n-1}\,(n-1)!\,(ax+b)^{-n} a^n .$$

(Proved)

**Th^m:**

If $y = \sin(ax+b)$, then show that

$$y_n = a^n \sin\left(n \cdot \frac{\pi}{2} + ax + b\right).$$

**Proof:**

Given, $y = \sin(ax+b)$ ——— (1)

Diff (1) wrt $x$. we have

$$y_1 = a\cos(ax+b) = a\sin\left(\frac{\pi}{2} + ax + b\right)$$

Diff. again wrt $x$ we get.

$$y_2 = a^2 \cos\left(\frac{\pi}{2} + ax + b\right) = a^2 \sin\left(\frac{\pi}{2} + \frac{\pi}{2} + ax + b\right)$$

$$= a^2 \sin\left(2 \cdot \frac{\pi}{2} + ax + b\right)$$

Similarly,

$$y_3 = a^3 \cos\left(2 \cdot \frac{\pi}{2} + ax + b\right) = a^3 \sin\left(\frac{\pi}{2} + 2 \cdot \frac{\pi}{2} + ax + b\right)$$

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$$\Rightarrow y_3 = a^3 \sin\left(3 \cdot \frac{\pi}{2} + ax + b\right)$$

$$y_n = a^n \sin\left(n \cdot \frac{\pi}{2} + ax + b\right).$$

(Proved)

**Th^m:**

If $y = \cos(ax+b)$, then show that

$$y_n = a^n \cos\left(n \cdot \frac{\pi}{2} + ax + b\right).$$

**Proof:**

Given, $y = \cos(ax+b)$ ——— (1)

Diff (1) wrt $x$ we get.

$$y_1 = -a \sin(ax+b) = a \cos\left(\frac{\pi}{2} + ax + b\right)$$

Again diff wrt $x$ we have

$$y_2 = -a^2 \sin\left(\frac{\pi}{2} + ax + b\right) = a^2 \cos\left(\frac{\pi}{2} + \frac{\pi}{2} + ax + b\right)$$

$$\Rightarrow y_2 = a^2 \cos\left(2 \cdot \frac{\pi}{2} + ax + b\right)$$

Similarly, $y_3 = -a^3 \sin\left(2 \cdot \frac{\pi}{2} + ax + b\right)$

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$$\Rightarrow y_3 = a^3 \cos\left(\frac{\pi}{2} + 2\cdot\frac{\pi}{2} + ax + b\right)$$

$$\Rightarrow y_3 = a^3 \cos\left(3\cdot\frac{\pi}{2} + ax + b\right)$$

- - -

$$y_n = a^n \cos\left(n\cdot\frac{\pi}{2} + ax + b\right).$$

(Proved)

**Th^m:**

If $y = e^{ax}\sin(bx+c)$, then show that

$$y_n = (a^2+b^2)^{n/2} e^{ax} \sin\left(bx + c + n\tan^{-1}\frac{b}{a}\right)$$

**Proof:**

Given, $y = e^{ax}\sin(bx+c)$ ——— (1)

Differentiating (1) wrt $x$ we get

$$y_1 = ae^{ax}\sin(bx+c) + be^{ax}\cos(bx+c)$$

$$\Rightarrow y_1 = e^{ax}\left[a\sin(bx+c) + b\cos(bx+c)\right] \quad \text{——— (2)}$$

Let, $a = r\cos\theta$ & $b = r\sin\theta$

Hence, $a^2 + b^2 = r^2(\cos^2\theta + \sin^2\theta) = r^2$

$\Rightarrow r = (a^2+b^2)^{1/2}$

& $\frac{b}{a} = \tan\theta$

$\Rightarrow \theta = \tan^{-1}\left(\frac{b}{a}\right)$

Now, from equⁿ ② we get

$$y_1 = e^{ax}\left[r\cos\theta \sin(bx+c) + r\sin\theta\cos(bx+c)\right]$$

$$= re^{ax}\left[\cos\theta\sin(bx+c) + \sin\theta\cos(bx+c)\right]$$

$$\therefore y_1 = re^{ax}\sin(bx+c+\theta)$$

Similarly, differentiating $y_1$ wrt $x$ we obtain

$$y_2 = r^2 e^{ax}\sin(bx+c+2\theta)$$

Proceeding in the similar manner, we have

$$y_n = r^n e^{ax}\sin(bx+c+n\theta)$$

$$\therefore y_n = (a^2+b^2)^{n/2} e^{ax}\sin\left(bx+c+n\tan^{-1}\left(\frac{b}{a}\right)\right)$$

(Proved)

**Th$^m$:**

If $y = e^{ax}\cos(bx+c)$, then show that

$$y_n = (a^2+b^2)^{n/2} e^{ax} \cos\left(bx+c+n\tan^{-1}\frac{b}{a}\right).$$

**Proof:**

Given, $y = e^{ax}\cos(bx+c)$ ——— (1)

Differentiating (1) wrt $x$ we have

$$y_1 = ae^{ax}\cos(bx+c) - be^{ax}\sin(bx+c)$$

$$\Rightarrow y_1 = e^{ax}\left[a\cos(bx+c) - b\sin(bx+c)\right] \quad \text{——— (2)}$$

Let, $a = r\cos\theta$ & $b = r\sin\theta$

so that $a^2+b^2 = r^2 \Rightarrow r = (a^2+b^2)^{1/2}$

$$\tan\theta = \frac{a}{b} \Rightarrow \theta = \tan^{-1}\left(\frac{a}{b}\right)$$

Hence, from (2) we have

$$y_1 = e^{ax}\left[r\cos\theta\cos(bx+c) - r\sin\theta\sin(bx+c)\right]$$

$$\Rightarrow y_1 = re^{ax}\left[\cos\theta\cos(bx+c) - \sin\theta\sin(bx+c)\right]$$

$$\Rightarrow y_1 = re^{ax}\cos(bx+c+\theta)$$

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Similarly, differentiating $y_1$ wrt $x$ we have

$$y_2 = r^2 e^{ax} \cos(bx+c+2\theta)$$

Proceeding in a similar manner, we obtain

$$y_n = r^n e^{ax} \cos(bx+c+n\theta)$$

$$\therefore y_n = (a^2+b^2)^{n/2} e^{ax} \cos(bx+c+n\tan^{-1}(b/a))$$

(Proved)

**Example:**

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If $y = x^{2n}$ then show that $y_n = 2^n\{1.3.5\cdots(2n-1)\}x^n$ where $n \in \mathbb{N}$.

**Sol$^n$:**

Given, $y = x^{2n}$ ——— (1)

Diff (1) wrt $x$ we have

$$y_1 = 2nx^{2n-1}$$

Again diff wrt $x$ we have

$$y_2 = 2n(2n-1)x^{2n-2}$$

$$y_3 = 2n(2n-1)(2n-2)x^{2n-3}$$

$$y_4 = 2n(2n-1)(2n-2)(2n-3)x^{2n-4}$$

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$$y_n = 2n(2n-1)(2n-2)(2n-3)\cdots\{2n-(n-1)\}x^{2n-n}$$

$$= 2n(n-1)(2n-2)(2n-3)\cdots(n+1)x^n$$

$$= \frac{2n(2n-1)(2n-2)\cdots(n+1)\cdot n!}{n!}x^n$$

$$\therefore y_n = \frac{(2n)!}{n!}x^n \qquad \text{——— (2)}$$

Now,

$$(2n)! = 2n(2n-1)(2n-2)(2n-3)(2n-4)\cdots 6.5.4.3.2.1$$

$$= \{2n(2n-2)(2n-4)\cdots 6.4.2\}\{(2n-1)(2n-3)\cdots 5.3.1\}$$

$$= 2^n\{n(n-1)(n-2)\cdots 3.2.1\}\{(2n-1)(2n-3)\cdots 5.3.1\}$$

$$= (2^n \cdot n!)\cdot\{(2n-1)(2n-3)\cdots 5.3.1\}$$

$$\Rightarrow \frac{(2n)!}{n!} = 2^n\{(2n-1)(2n-3)\cdots 5.3.1\} \qquad \text{——— (3)}$$

Now, using (3) equⁿ (2) can be rewritten as

$$y_n = 2^n\{1.3.5\cdots(2n-3)(2n-1)\}x^n \quad ; \quad n \in \mathbb{N}.$$

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**Problem:**

If $y = \sin(\sin x)$, then show that

$$\frac{d^2y}{dx^2} + \tan x \frac{dy}{dx} + y\cos^2 x = 0$$

**Sol$^n$:**

Given, $y = \sin(\sin x)$ ——— (1)

Diff. (1) wrt $x$ we have

$$\frac{dy}{dx} = \cos(\sin x)\cos x \quad \text{——— (2)}$$

Again diff (2) wrt $x$ we have

$$\frac{d^2y}{dx^2} = -\sin(\sin x)\cos^2 x - \cos(\sin x)\sin x$$

$$= -y\cos^2 x - \frac{1}{\cos x}\frac{dy}{dx}\cdot \sin x \quad [\text{using (1) \& (2)}]$$

$$\Rightarrow \frac{d^2y}{dx^2} = -y\cos^2 x - \tan x \frac{dy}{dx}$$

$$\Rightarrow \frac{d^2y}{dx^2} + \tan x \frac{dy}{dx} + y\cos^2 x = 0$$

(Showed)

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**Problem:**

If $y = \ln(x + \sqrt{a^2+x^2})$ then show that

$(a^2+x^2)y_2 + xy_1 = 0$

**Soln:**

Given, $y = \ln(x + \sqrt{a^2+x^2})$ ——— (1)

Diff (1) wrt $x$ we have

$$y_1 = \frac{1}{x+\sqrt{a^2+x^2}}\left(1 + \frac{1}{2\sqrt{a^2+x^2}} \cdot 2x\right)$$

$$= \frac{1}{x+\sqrt{a^2+x^2}}\left(1 + \frac{x}{\sqrt{a^2+x^2}}\right)$$

$$= \frac{1}{(x+\sqrt{a^2+x^2})}\left[\frac{(x+\sqrt{a^2+x^2})}{\sqrt{a^2+x^2}}\right]$$

$$\Rightarrow y_1 = \frac{1}{\sqrt{a^2+x^2}} = (a^2+x^2)^{-1/2} \quad \text{——— (2)}$$

Again diff (2) wrt $x$ we obtain

$$y_2 = -\frac{1}{2}(a^2+x^2)^{-3/2} \cdot 2x$$

$$\Rightarrow y_2 = -x(a^2+x^2)^{-1} \cdot (a^2+x^2)^{-1/2}$$

$$\Rightarrow (a^2+x^2)y_2 = -x(a^2+x^2)^{-1/2}$$

$$= -xy_1 \quad [\text{By (2)}]$$

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$\Rightarrow (a^2+x^2)y_2 + xy_1 = 0$

(Showed)

**Problem:**

If $y = a\cos(\ln x) + b\sin(\ln x)$ then show that

$$x^2y_2 + xy_1 + y = 0$$

**Sol$^n$:**

Given, $y = a\cos(\ln x) + b\sin(\ln x)$ ——— (1)

Diff (1) wrt $x$ we have

$$y_1 = -a\sin(\ln x)\cdot\frac{1}{x} + b\cos(\ln x)\cdot\frac{1}{x}$$

$$\Rightarrow xy_1 = -a\sin(\ln x) + b\cos(\ln x) \quad \text{——— (2)}$$

Again diff (2) wrt $x$ we have

$$y_1 + xy_2 = -a\cos(\ln x)\cdot\frac{1}{x} - b\sin(\ln x)\cdot\frac{1}{x}$$

$$\Rightarrow xy_1 + x^2y_2 = -[a\cos(\ln x) + b\sin(\ln x)]$$

$$\Rightarrow x^2y_2 + xy_1 = -y \quad [\text{using (1)}]$$

$$\Rightarrow x^2y_2 + xy_1 + y = 0$$

(Showed)

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**Problem:**

If $y = \sin(m \sin^{-1} x)$ then show that

$$(1-x^2) y_2 - x y_1 + m^2 y = 0.$$

**Soln:**

Given, $y = \sin(m \sin^{-1} x)$ ——— (1)

Diff (1) wrt $x$ we have

$$y_1 = \cos(m \sin^{-1} x) \cdot \frac{m}{\sqrt{1-x^2}}$$

$$\Rightarrow y_1 \sqrt{1-x^2} = m \cos(m \sin^{-1} x) \quad \text{——— (2)}$$

Again diff (2) wrt $x$ we obtain

$$y_2 \sqrt{1-x^2} + y_1 \frac{1}{2\sqrt{1-x^2}} (-2x) = -m \sin(m \sin^{-1} x) \cdot \frac{m}{\sqrt{1-x^2}}$$

$$\Rightarrow y_2 (1-x^2) - x y_1 = -m^2 \sin(m \sin^{-1} x)$$

$$\Rightarrow (1-x^2) y_2 - x y_1 + m^2 y = 0 \quad [\text{using (1)}]$$

(Showed)

**Problem:**

If $\ln y = \tan^{-1} x$ then show that

$$(1+x^2) y_2 + (2x-1) y_1 = 0$$

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Sol$^n$:

Given, $\ln y = \tan^{-1} x$ ——— (1)

Diff (1) wrt $x$ we have

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{1}{1+x^2}$$

$$\Rightarrow (1+x^2) y_1 = y \quad \text{——— (2)}$$

Again diff (2) wrt $x$ we obtain

$$(1+x^2) y_2 + 2x y_1 = y_1$$

$$\Rightarrow (1+x^2) y_2 + (2x-1) y_1 = 0$$

(Showed)

Problem:

If $y = e^{\cot^{-1} x}$ then show that $(1+x^2) y_2 + (2x+1) y_1 = 0$

Sol$^n$:

Given, $y = e^{\cot^{-1} x}$ ——— (1)

Diff (1) wrt $x$ we obtain

$$y_1 = e^{\cot^{-1} x} \left( -\frac{1}{1+x^2} \right)$$

$$\Rightarrow (1+x^2) y_1 = -e^{\cot^{-1} x} = -y \quad [\text{using (1)}]$$

$$\Rightarrow (1+x^2) y_1 + y = 0 \quad \text{——— (2)}$$

Again diff (2) wrt $x$ we obtain

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$$(1+x^2)y_2 + 2xy_1 + y_1 = 0$$

$$\Rightarrow (1+x^2)y_2 + (2x+1)y_1 = 0$$

(Showed)

**Problem:**

If $y = (\sin^{-1}x)^2$ then show that $(1-x^2)y_2 - xy_1 - 2 = 0$

**Soln:**

Given, $y = (\sin^{-1}x)^2$ ——— (1)

Diff (1) wrt $x$ we obtain

$$y_1 = 2\sin^{-1}x \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow y_1\sqrt{1-x^2} = 2\sin^{-1}x \quad \text{——— (2)}$$

Again diff (2) wrt $x$ we have

$$y_2\sqrt{1-x^2} + y_1 \frac{1}{2\sqrt{1-x^2}}(-2x) = 2 \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow y_2(1-x^2) - xy_1 = 2$$

$$\Rightarrow (1-x^2)y_2 - xy_1 - 2 = 0$$

(Showed)

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**Problem:**

If $y=(x^2-1)^n$ then show that $(x^2-1)y_2+2(1-n)xy_1-2ny=0$

**Sol$^n$:**

Given, $y=(x^2-1)^n$ ——— ①

Diff ① wrt $x$ we obtain

$$y_1 = n(x^2-1)^{n-1}\cdot 2x$$

$$\Rightarrow y_1 = \frac{2nx(x^2-1)^n}{(x^2-1)}$$

$$\Rightarrow (x^2-1)y_1 = 2nx(x^2-1)^n = 2nxy \quad [\text{using } ①]$$ ——— ②

Again diff ② wrt $x$ we have

$$(x^2-1)y_2 + 2xy_1 = 2ny + 2nxy_1$$

$$\Rightarrow (x^2-1)y_2 - 2nxy_1 + 2xy_1 - 2ny = 0$$

$$\Rightarrow (x^2-1)y_2 - 2(n-1)xy_1 - 2ny = 0$$

$$\Rightarrow (x^2-1)y_2 + 2(1-n)xy_1 - 2ny = 0$$

(Showed)

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**Problem:**

If $y = e^{a\sin^{-1}x}$ then show that $(1-x^2)y_2 - xy_1 - a^2y = 0$

**Sol$^n$:**

Given, $y = e^{a\sin^{-1}x}$ ——— (1)

Diff (1) wrt $x$ we get.

$$y_1 = e^{a\sin^{-1}x} \cdot \frac{a}{\sqrt{1-x^2}}$$

$$\Rightarrow \sqrt{1-x^2}\, y_1 = ae^{a\sin^{-1}x} = ay \quad \text{——} (2) \quad [\text{using } (1)]$$

Again diff (2) wrt $x$ we obtain

$$\left(\sqrt{1-x^2}\right)y_2 + y_1 \cdot \frac{1}{2\sqrt{1-x^2}} \cdot (-2x) = ay_1$$

$$\Rightarrow (1-x^2)y_2 - xy_1 = a\, y_1 \sqrt{1-x^2}$$

$$\Rightarrow (1-x^2)y_2 - xy_1 = a \cdot ay \quad [\text{using } (2)]$$

$$\Rightarrow (1-x^2)y_2 - xy_1 = a^2y$$

$$\Rightarrow (1-x^2)y_2 - xy_1 - a^2y = 0$$

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**Problem:**

If $y = \sin(m \sin^{-1} x)$ then show that

$$(1-x^2) y_3 - 3x y_2 + (m^2 - 1) y_1 = 0$$

**Soln:**

Given, $y = \sin(m \sin^{-1} x)$ ——— (1)

Diff (1) wrt $x$ we have

$$y_1 = \cos(m \sin^{-1} x) \cdot \frac{m}{\sqrt{1-x^2}}$$

$$\Rightarrow y_1 \sqrt{1-x^2} = m \cos(m \sin^{-1} x) \quad \text{——— (2)}$$

Again diff (2) wrt $x$ we obtain

$$y_2 \sqrt{1-x^2} + y_1 \frac{1}{2\sqrt{1-x^2}} (-2x) = -m \sin(m \sin^{-1} x) \cdot \frac{m}{\sqrt{1-x^2}}$$

$$\Rightarrow (\sqrt{1-x^2}) y_2 - x y_1 \cdot \frac{1}{\sqrt{1-x^2}} = \frac{-m^2 y}{\sqrt{1-x^2}}$$

$$\Rightarrow (1-x^2) y_2 - x y_1 = -m^2 y \quad \text{——— (3)}$$

Diff (3) wrt $x$ gives

$$(1-x^2) y_3 - 2x y_2 - y_1 - x y_2 = -m^2 y_1$$

$$\Rightarrow (1-x^2) y_3 - 3x y_2 + (m^2 - 1) y_1 = 0$$ (shown)

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**Problem:**

If $y = x^2 \ln x$ then show that $xy_3 = 2$.

**Soln:**

Given, $y = x^2 \ln x$ ——— ①

Diff ① wrt $x$ we obtain.

$$y_1 = 2x \ln x + x^2 \cdot \frac{1}{x}$$

$$\Rightarrow y_1 = 2x \ln x + x \quad \text{——— ②}$$

Again diff ② wrt $x$ we have

$$y_2 = 2x \cdot \frac{1}{x} + 2 \ln x + 1$$

$$\Rightarrow y_2 = 2 + 2 \ln x + 1 = 2 \ln x + 3 \quad \text{——— ③}$$

Again diff ③ wrt $x$ we get.

$$y_3 = 2 \cdot \frac{1}{x}$$

$$\Rightarrow xy_3 = 2$$

(Showed)

**Problem:**

If $\ln y = \tan^{-1} x$ then show that

$$(1+x^2) y_3 + (4x-1) y_2 + 2y_1 = 0$$

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Soln:

Given, $\ln y = \tan^{-1} x$ ——— (1)

Diff (1) wrt $x$ we get

$$\frac{1}{y} y_1 = \frac{1}{1+x^2}$$

$$\Rightarrow (1+x^2) y_1 = y \quad \text{——— (2)}$$

Diff (2) wrt $x$ we obtain

$$(1+x^2) y_2 + 2x y_1 = y_1$$

$$\Rightarrow (1+x^2) y_2 + (2x-1) y_1 = 0 \quad \text{——— (3)}$$

Diff (3) wrt $x$ we have

$$(1+x^2) y_3 + 2x y_2 + 2 y_1 + (2x-1) y_2 = 0$$

$$\Rightarrow (1+x^2) y_3 + (4x-1) y_2 + 2 y_1 = 0$$

(Showed)

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